MILITARY SPECIFICATION

GUIDED WEAPON CONTROL
GROUP OK-293A/AWW

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the OK-293A/AWW Control Group, Guided Weapon, herein after referred to as the control group. The control group consists of Control, Guided Weapon C-9743A/AWW (Joystick) and Converter, Control Signals C-3248A Electronic Package). The control group is to be located in the aircraft and is to interface with the Guided Weapon Control-MWtm Sets AN/AWW-7B, or AN/AWW-9 (Pod) to provide post-launch cockpit control for retargeting and steering of the WALLEYE. Data link Guided Weapon. The control group also provides proportional dc voltage inputs to the AN/ASN-91 computer.

1.2 Associated equipment. The control group operates with the associated equipment listed steal in 6.4.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, and standards of the issue listed in that issue of Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1430

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SPECIFICATIONS

FEDERAL

PPP-T-60	Tape, Packaging, Waterproof
PPP-T-76	Tape, Pressure Sensitive Adhesive paper {For Carton Sealing)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard
QQ-S-781	Strapping, Steel, and Seals

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life, for Aircraft Electronic Equipment, Format For
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Support Equipment
MIL-H-46855	Human Engineering Requirements for Military System Equipment and Facilities

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads (40″ x 48″ 4 Way Partial and 4 Way Pallets)
MIL-STD-454	Standard General Requirements for Electronic Equipment

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MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilizat{on of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-965	Parts Control Program
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities

2.1.2 Other Government documents, drawing, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

Naval Air Systems Command

NAVAIR 16-350K293-1	Guided Weapon Control Group OK-293A/AWW

DRAWING

Naval Afr Systems Command

DL 475AS150	Data List for Guided Weapon Control Group OK-293A/AWW

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connections with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specWicatfon shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

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3.2 Parts and materials.

3.2.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-E-5400.

3.2.2 Basic conformance. Parts and materials selected for use in the equipment shall be in accordance with MIL-E-5400. Parts shall be selected from those conforming to MIL-STD-965, unless the parts selected will not perform as required in the intended applications and environments specified herein in which case 3.2.1 shall be used to select and obtain approval for nonstandard parts.

3.2.3 Interchangeability. The control group shall meet the interchange-abillty requirements of MIL-E-5400 Drawing DL 475AS150 shall be the standard for measuring interchangeability of all applicable units, assemblies. subassemblies, and parts.

3.3 Design and construction. The control group shall be manufactured in accordance with DL 475AS150 except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this specifica tion. In any case it is a condition of final acceptance that the control group manufactured meet all the performance requirements of this specification.

3.3.1 Dimensions. The dimensions of the control group shall be not greater than specified in table I.

3.3.2 Total weight. Total weight of control group shall be not greater than 3.18 Pounds (1.44 kilograms).

3.3.3 Reliability.

3.3.3.1. Operational stability. The control group shall meet the performance requirements of th is specification when operated continuously or intermittently for a period of at least one MTBF (752 hours) wfthout the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.3.2 Operating life. The control group shall have a total operating life of 5,000 hours with a minimum servicing and replacement of parts.

3.3.3.3 Reliability in Mean-Time-Between-Failure (MTBF). The control group shall have a minimum of 752 hours or mean (operating) time between failures when tested and accepted as specified in 4.5.3.

3.3.4 Maintainability Maintainability provisions shall beincorporated within the design to provide for service, access and repair of all assemblies. Normal routine testing of the control group shall be accomplished using Government furnished Distribution Box J-3416A/AWW (see 3.6) and the standard test equipment normally available in intermediate maintenance shops.

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3.3.5 Transportability. The control group shall be transportable by air, rail, truck and ship when mounted in a shipping container as specified in Section 5.

TABLE I. Dimensions

Components	Length	Width	Height
Joystick	8.375 inches	3.75 inches	2.25 inches
Electronics Package	5.0 inches	4.5 inches	5.25 inches

3.3.6 Nameplates and identification marking. Serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

3.3.6.1 Identification plates. Identification plates for the joystick and electronics package shall in accordance with DL 475AS150.

3.3.6.2 Decals, nameplates and markings locations. All decals, namepleates, and markings shall be affixed to the control group in accordance with DL 475AS150.

3.3.6.3 Serial numbers. Serial numbers will be assigned by the procuring activity for the control group specified in the contract or purchase order. The contractor shall serialize sequentially all control groups which are specified in the contract.

3.3.7 Color and finish. Color and finish of the control group shall be in accordance with DL 475AS150 and MIL-E-5400

3.3.8 Interference control. The generation of electromagnetic interference by the control group and the vulnerability of the control group to electromagnetic interference shall be controlled within the limits of MIL-STD-461 for CLass A1 equipment. EMI testing shall be accomplished under standard conditions specified in 3.3.10 and in accordance with the procedures of MIL-STD-462 to demonstrate compliance with the MIL-STD-461.

3.3.9 Safety. Safety requirements shall be in accordance with MIL-E-5400.

3.3.9.1 Human performance/Human engineering. The control group and all component parts thereof, shall be designed in accordance with MIL-STD-1472. Recognized principles of human engineering as specified in MIL-H-46855 shall be followed during design.

3.3.10 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a. Temperature — Room ambient (25 ±5°C)

b. Altitude — Normal ground

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c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative
e.	Input power voltage	+28 ± .1 VDC, +4 ±0.01 VDC, -4 ±.01 VDC,+5 ±0.1 VDC (Panel Voltage)

3.3.11 Service conditions. The control group shall meet the performance requirements of this specification under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for class 1 equipment, except as modified herein.

3.3.11.1 Vibration. The control group shall meet the performance requirements of this specification during and after being subjected to the vibration environment specified in 4.5.1. The performance characteristics specified in 4.5.1.3 shall apply.

3.3.11.2 Temperature-altitude. The control group shall meet the temperature-altitude combination requirements of MIL-E-5400 for Class 1 equipment.

3.3.11.3 Shock.

3.3.11.3.l Operational. The control group shall meet the performance requirements after shock testing in accordance with MIL-E-5400 except that the amplitude for the joystick shall be 20g in longitudinal axis, 22g in the lateral axis, and 35g in the vertical axis. The amplitudes for the electronic package shall be 20g in longitudinal axis, 30g in the lateral axis, and 46g in the vertical axis. Figure 5 specifies the appropriate axes.

3.3.11.3.2 Crash safety. The control group shall meet the requirements of MIL-E-5400 except that the amplitudes for the joystick shall be 40g longitudinal, 40g lateral, and 47g vertical. The amplitudes for the electronic package shall be 40g longitudinal, 40g lateral, and 55g vertical.

3.3.11.4 Humidity. The control group shall meet the performance requirements of this specification when subjected to the humidity exposure in accordance with MIL-T-5422.

3.3.11.5 Salt spray. The control group shall meet the performance requirements of this specification when subjected to the salt spray exposure in accordance with MIL-T-5422.

3.3.11.6 Explosion. The control group shall meet the performance requirements of this specification when subjected to the explosion tests in accordance with MIL-T-5422.

3.3.12 Warm-up time. There is no warm-up time required for the control group prior to operation when the temperature is from +54°C to -20°C. At temperatures below -20°C, three minutes warm-up is required.

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3.3.13 Electrical power inputs, grounds and loads.

3.3.13.1 Operating power. The control group shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The input power required to operate the controller shall be as follows:

a. +28.0 ±0.1 volts dc to pin H of J8 (category A, 75 watts)

b. +5.0 ±0.1 volts dc to pin D of J102 (1.5 watts)

c. +4.0 ±0.01 volts dc to pin G of J101 (24 mw)

d. -4.0 ±0.01 volts dc to pin F of J101 (24 mw)

3.3.13.2 Ground and load conditions. The input power (see 3.3.13.1) requires certain ground and load conditions as follows:

a. Ground pin F of J8

b. 100 ohms ±1 percent from pin A of J8 to ground

c. Ground pin J of J101

3.3.13.3 Interface connections. See figure 1 for the interface connections of the joystick and electronics package.

3.4 Performance, The performance of the control group shall be as specified below any and all performance requirements specified on DL 475AS150. (Detail funciton and description for the control group is covered in NAVAIR 16-350K293-1.) Unless otherwise specified herein, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations shall be as specified below:

a. Under standard conditions (see 3.3.10), the control group shall provide the regulated outputs as specified in table II.

b. Under service conditions (see 3.3.11), the control group shall provide the adjusted and regulated outputs specified in table III.

c. The resistance between J102-F and J101-J on the joystick assembly shall be 10 megohms minimum.

3.5 Detail requirements.

3.5.1 Control, Guided Weapon C-9743A/AWW (joystick). The joystick shall be manufactured in accordance with DL 475AS150 and the requirements specified in 3.5.1.1 through 3.5.1.4.4.

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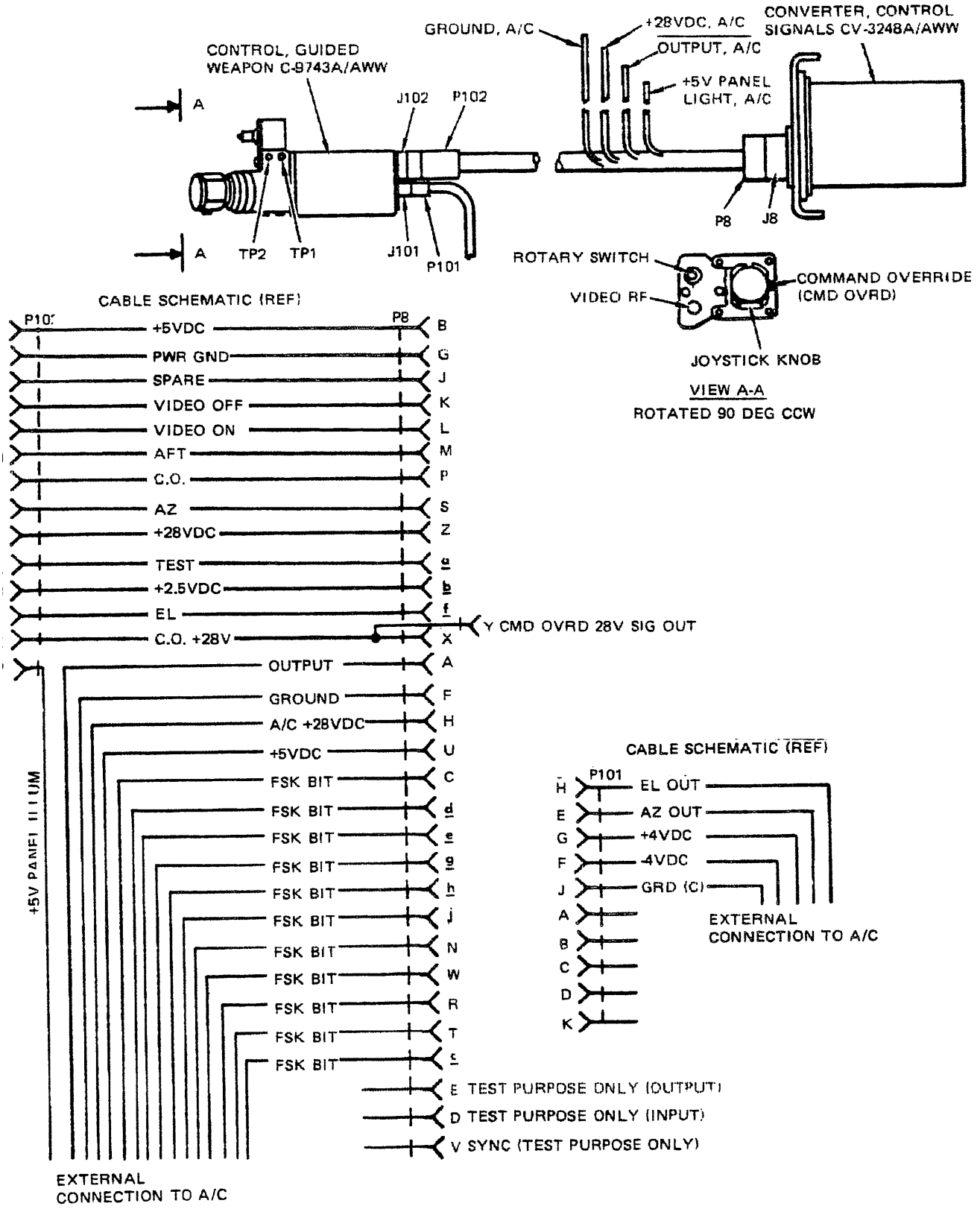


FIGURE 1. Electrical schematic showing interface connection between joystick and electronic package.

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3.5.1.1 Joystick function. The Control, Guided Weapon C-9743A/AWW shall provide fIve commands OF 5 volts DC or 0 volt DC for an on/off condition and two DC voltages (azimuth and elevation) that vary between 0 volt DC and 5 volts DC to voltage controlled oscillators (VCO) in the convertor, control signal CV-3248A/AWW. The joystick shall provide proportional azimuth and elevation dc voltage inputs to the AN/ASN-91 computer. The joystick control panel shall contain a four position mode selector switch, a video rf switch, a stiff force-proportional joystick, and the Command Override push button switch in the joystick handle. The mode selector, video rf, and command override switches shall activate tone oscillators which produce a frequency for each discrete command sent. The force-proportional joystick shall control two signal channels, azimuth and elevation; each of which shall put out an amplified proportional dc voltage to a voltage controlled oscillator (VCO).

3.5.1.2 Joystick weight. The weight of the joystick shall be not greater than 1.41 pounds.

3.5.1.3 Joystick content. The joystick shall contain the following circuits:

a. Weapon Circuit (A1 Board) The A1 board shall provide two DC voltages (azimuth and elevation) which vary between O volt DC and 5 volts DC and one 5 volts DC or 0 volt DC voltage for an on/off condition. These voltages are the inputs to Voltage controlled Oscillators (VCO). The A1 board shall be manufactured in accordance with DL 475AS150.

b. Computer Circuit (A2 Board) - The A2 board shall provide two dc voltages (azimuth and elevation) which vary between a +4 volts dc and -4 volts dc to the Navigational Weapon Delivery System (NWDS). It also shall insure that a 0.00 ±0.148 volts dc output exists around the joystick center position. The A2 board shall remanufactured in accordance wtth DL 475AS150.

3.5.1.4 Joystick controls.

3.5.1.4.1. Mode selector switch (front panel). The mode selector switch Shall enable the weapon video transmitter to transmit rf power in the low power mode and shall provide for selection of either the pod forward (FWD) or AFT antenna. Computer, AN/ASN-91, receives input signals when the joystick is in normal mode.

3.5.1.4.2 Video rf switch (front panel). The video rf switch shall provide, via data link, prelaunch ON-OFF control of the video transmitter in the weapon, shall initiate video tape rwmkr tape movement in the pod and lights a weapon readiness indicator on the pod station.

3.5.1.4.3 Command override button (on the side of joystick handle). This control shall cause the weapon to “break” target “lock” when pressed and shall lock-on to another target when button is released.

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TABLE II. Output requirements, standard conditions

Joystick Switch Position	Joystick Displacement	OUTPUTS (J8-A) Freq (KHz)	(J8-A) Volts (RMS)	(J101-E) AZ VOLT DC	(J101-H) EL VOLT DC
ANT FWD	NONE	30 ±0.06	117 ±6 MV	-	-
Depress Command Override	Full FWD	27.75 ±0.45	-	-	-
Depress Command Override	Full AFT	32.250 ±0.45	-	-	-
-	None	22 ±0.05	70 ±4 MV	-	-
Depress Command Override	Full Left	23.65 ±0.33	-	-	-
Depress Command Override	Full Right	20.35 ±0.33	-	-	-
Depress Command Override	None	10.5 ±0.03	27 ±2.5 MV	-	-
Video RF to OFF	None	7.35 ±0.02	17.5 ±2 MV	-	-
Video RF to ON	None	5.4 ±0.02	12 ±2 MV	-	-
Test	None	1.7 ±0.01	10 ±1.5 MV	-	-
AFT ANT	None	1.3 ±0.005	10 ±1.5 MV	-	-
NORMAL	None	-	-	0 ±0.074	0 ±0.074
NORMAL	Full FWD	-	-	-	-3.9 +0.05 -0.10
NORMAL	Full AFT	-	-	-	+3.9 +0.10 -0.05
NORMAL	Full Left	-	-	-3.9 +0.05 -0.10	-
NORMAL	Full Right	-	-	+3.9 +0.10 -0.05	-
FWD ANT	All Directions	-	-	0 ±0.005	0 ±0.005
AFT ANT	All Directions	-	-	0 ±0.005	0 ±0.005
Test	All Directions	-	-	0 ±0.005	0 ±0.005

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TABLE III. Output requirements, service condition

Joystick Switch Position	Joystick Displacement	OUTPUTS (J8-A) Freq (KHz)	(J8-A) Volts (MV)	(J101-E) AZ VOLT (DC)	(J101-M) EL VOLT (DC)
FWD ANT	None	30 ±0.196	117 ±18	-	-
Depress Command Override	Full FWD	27.75 ±0.45	-	-	-
Depress Command Override	Full AFT	32.250 ±0.45	-	-	-
-	None	22 ±0.150	70 ±11 MV	-	
Depress Command Override	Full Left	23.65 ±0.33		-	
Depress Command Override	Full Right	20.35 ±0.33	-	-	
Depress Command Override	None	10.5 ±0.078	27 ±6	-	
Video RF to OFF	None	7.35 ±0.054	17.5 ±4		
Video RF to ON	None	5.4 ±0.044	12 ±4	-	
Test	None	1.7 ±0.018	10 ±3	-	-
AFT ANT	None	1.3 ±0.016	10 ±3	-	-
NORMAL	None	-	-	0 ±0.148	0 ±0.148
NORMAL	Full FWD				+3.9 +0.05 -0.10
NORMAL	Full AFT	-	-	-	+3.9 +0.10 -0.05
NORMAL	Full Left			-3.9 +0.05 -0.10	
NORMAL	Full Right	-	-	+3.9 +0.10 -0.05	
FWD ANT	All Directions	-	-	0 ±0.005	0 ±0.005
AFT ANT	All Directions	-	-	0 ±0.005	0 ±0.005
Test	All Directions	-	-	[illegible]	[illegible]

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3.5.1.4.4 Stiff force-proportional joystick. When the mode selector switch is in NORMAL position, two signal channels (Azimuth and Elevation} shall be provided to the Computer AN/ASN-91. Figure 2 specifies the signal output to the computer vs. force applied to the joystick. When the mode selector switch is in any position other than NORMAL, the two signal channels, (Azimuth and Elevation) each shall put out an amplified dc voltage proportional to the applied force. Figure 3 specifies the voltage output vs. force applied to the joystick.

3.5.1.5 Joystick electrical connections. Connections to external circuits shall be provided as speci fied in table IV.

TABLE IV. Joystick electrical connections

Ref. Des	Receptacle	Function
J101	MS3112E12-10P	Interfaces with Computer AN/ASN-91
J102	475AS913	Interfaces with converter

3.5.2 Converter, Control Signal CV-3248A/AWW (electronics package). The electronic package shall be manufactured in accordance with 475AS150 and the requirements specified in 3.5.2.1 through 3.5.2.4.

3.5.2.1 Electronics package function. The electronics package shall provide regulated operating voltages for the control group. It shall contain the voltage controlled oscillators (VCO) and associated electronic switches and constant current circuitry. It also shall contain two thermal heaters and a thermal switch to maintain a stable operating environment for the VCO'S at temperature below -20°C.

3.5.2.2 Electronics package weight. The weight of the electronics package shall be not greater than 1.76 pound (798.3 grams).

3.5.2.3 Electronic package content. The electronics package shall contain the following circuits.

a . Electronic Package (2A1 Board) - The electronics package circuit board 2A1 shall provide +5 volts dc and 2.5 volts dc to the A1 board in the joystick. Also, the circuit board shall convert dc voltages into ac signals that are sinned together and then converted into a constant current source. The 2A1 board. shall be manufactured in accordance with IX 475AS150.

b. Electronic Package (2A2 Board) - The electronics package circuit board 2A2 shall provide a 16 bit code with two bits sync and 14 bits data. The first half of each bit and the first two bits shall be +2.5 V dc. The next 12 bits shall be 0 V dc or +5 V dc depending on the thumbwheel switch

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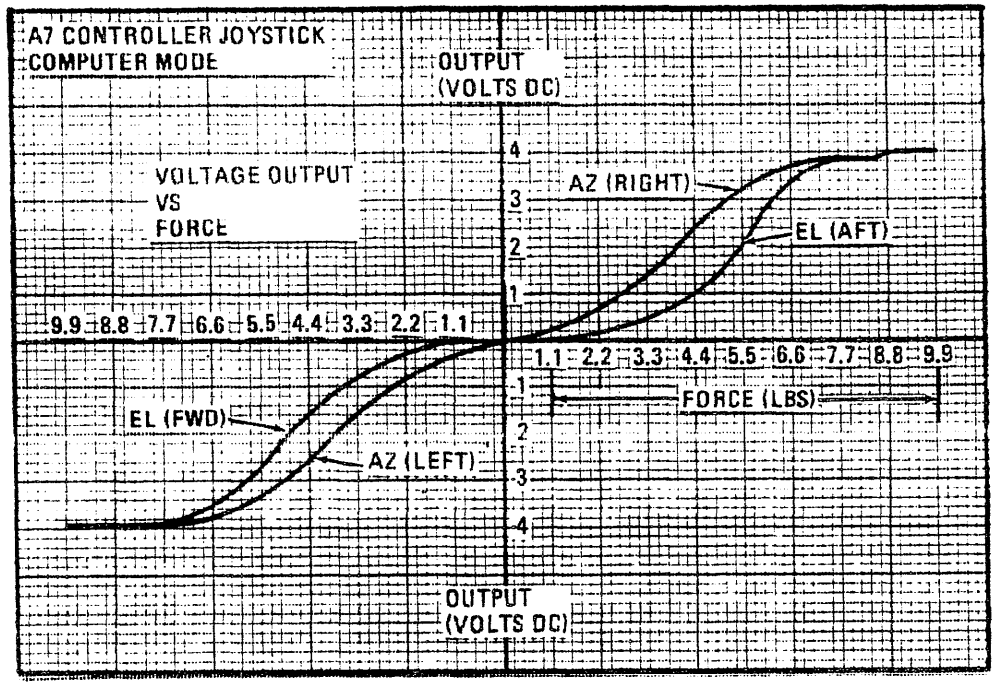


FIGURE 2. Output signal vs Force applied to the joystick.

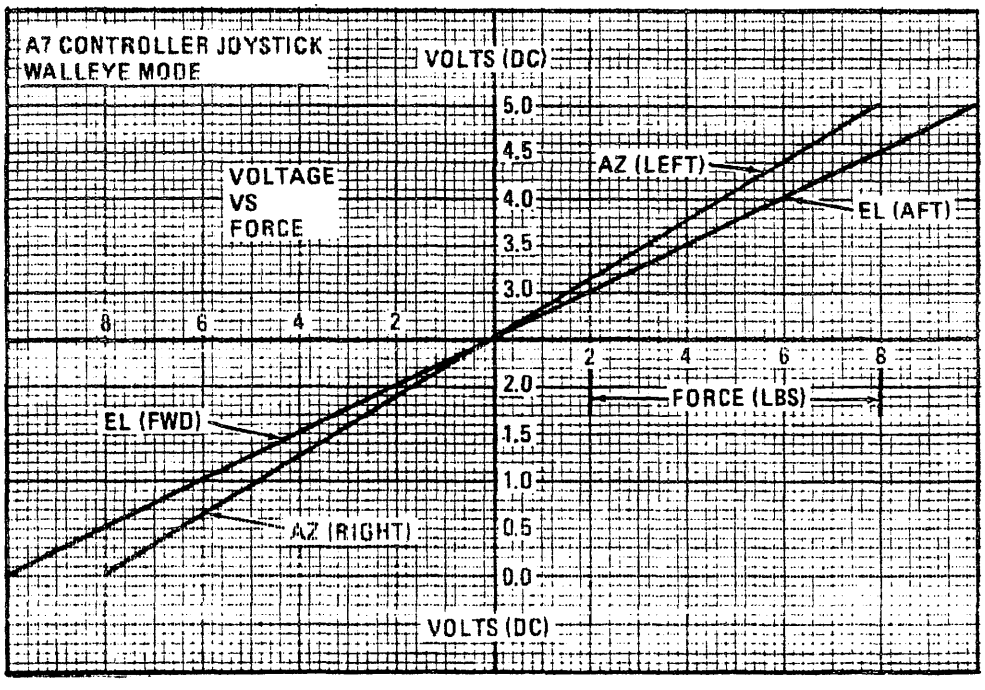


FIGURE 3. DC signal to the voltage controlled oscillators (VCO) vs force applied to the joystick.

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position on the frequency select panel in the cockpit. The last two bits shall be tied low. When the thumbwheel switch is placed in position M (manual), the board shall supply signals that are decoded by the AN/AWW-7B pod. The 2A2 board shall be manufactured in accordance with DL 475AS150.

c. The electronics package contains the Voltage Controlled Oscillators (VCO) listed in table V. The output of Y6 and Y7 shall be deviated around the center frequency in proportion to the force applied to the joystick as indicated in figure 4. The output of Y1 shall be deviated around the center frequency in proportion to the digital word applied from board 2A2.

d. Printed Circuit Board bracket Assembly (475AS831) - The bracket assembly shall provide +28 volts dc to the voltage control oscillators (VCO) listed in table V and to heaters 475AS916.

TABLE V. Voltage controlled oscillators frequency and function.

VCO	Freq (KHz)	Function
Y7	30.00	Elevation
Y6	22.00	Azimuth
Y5	10.50	Command Override
Y4	7.35	Video OFF
Y3	5.40	Video ON
Y2	1.70	Test
Y1	1.30	AFT

3.5.2.4 Electronics package electrical connections. Connections to external circuit shall be provided as listed in table VI.

TABLE VI. Electronics package electrical connections

Ref Des	Receptacle Type	Function
J8	MIL-C-83723/06-R1832N	Interfaces with Joystick and Frequency Select Panel

3.6 Government loaned property. A J-3416A/AWW Distribution Box will be furnished to the contractor for performance testing purposes.

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3.7 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contrtact or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to precribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

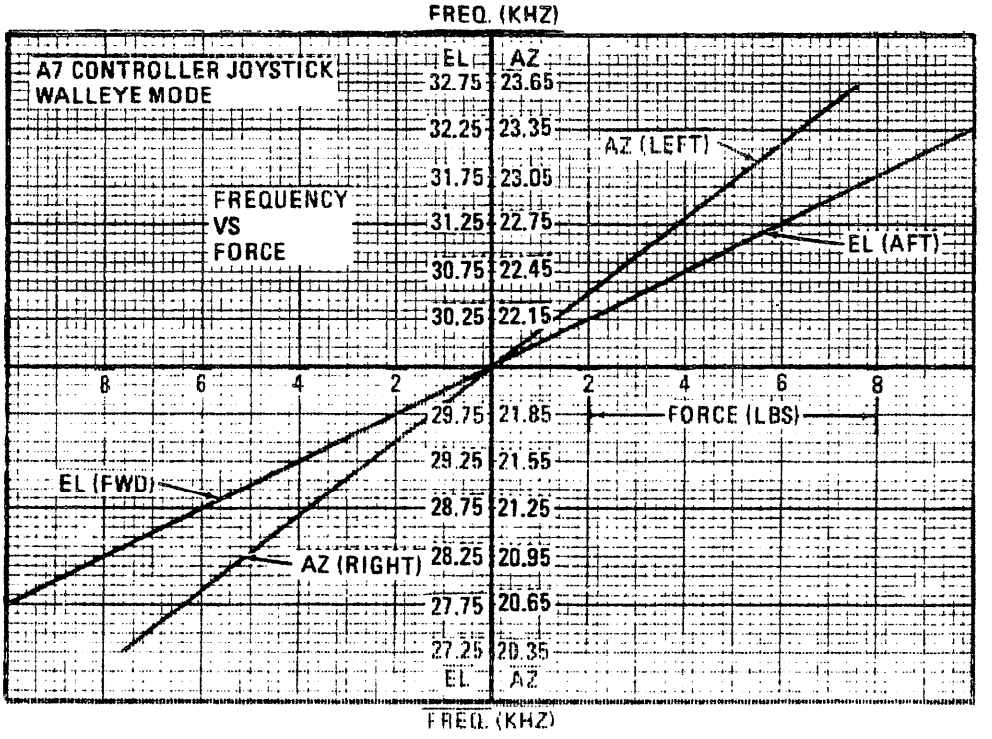


FIGURE 4. Output frequency of VCO vs force applied to the joystick

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4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 3.3.10 and 3.3.11.

4.4 First article inspection. First article inspection shall be conducted by the contractor on a control group representative of the production control groups to be supplied under the contract. First article tests shall be accomplished under the approved test procedures (see 4.9). Column I of table VII lists minimum tests required for first article inspection. No first article inspections shall be conducted prior to acceptance of first article test procedure by the procuring activity (see 6.2.2).

4.4.1 First article test data. Data collected in conducting ftrst article tests shall be made available to the procuring activity (see 6.2.2).

4.4.2 Scope of first article tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the control group meets all the requirements of this specification and the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-T-5422 and interference tests in accordance with MIL-STD-461.

4.4.3 First article approval. Approval of first article sample will be by the procuring activity upon completion of all tests and the first article test report being made available (see 6.2.2). No production control groups shall be accepted by the government prior to approval of the first article sample.

4.4.4 Production control groups. Control groups supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equal to the approved first article sample. Each control group shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection of a control group and of control groups already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.5 Quality conformance Inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspection. All inspection and testing shall be under the supervision of the government inspector. The contractor shall prepare and make available test reports showing quantitive results for all quality conformance tests (see 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspection shall consist of the following:

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TABLE VII. Inspection requirements

Requirement	I. First Article Inspection Paragraphs	II. Quality Conformance Inspections Paragraphs			
		Individual (A)	Sampling (B)	Special (C)	Life (D)
Visual Examination (see 3.2,3.3)	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1
Performance Characteristics (see 3.4)	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2
Workmanship (see 3.7)	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3
Vibration (see 3.3.11.1)	4.7.1				(see 4.5.6.2, and 4.5.6.3)
Temperature-Altitude (see 3.3.11.2)	4.7.2				
Shock (see 3.3.11.3)	4.7.3				
Humidity (see 3.3.11.4)	4.7.4				
Salt Spray (see 3.3.11.5)	4.7.5	1/	2/	3/	
Explosion (see 3.3.11.6)	4.7.6				
Warm-up time (see 3.3.12)	4.8				
Interference Control (see 3.3.8)	4.6				
Reliability (see 3.3.3)	4.5.3.1		4.5.3.2		

1/ Additional individual tests may be requested by the procuring activity (see 4.5.1).

2/ Sampling tests to be performed will be determined by the procuring activity based on first article and individual test results (see 4.5.2.1).

3/ Special tests to be performed will be determined by the procuring activity (see 4.5.4.2).

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a. Individual Tests (see 4.5.1}

b. Sampling Tests (see 4.5.2)

c. Reliability Assurance Tests (see 4.5.3)

d. Special Tests (see 4.5.4)

e. Life Tests (see 4.5.6)

4.5.1 Individual tests. Each control group submitted for quality conformance inspection shall be subjected to the individual tests. These tests shall determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each control group accepted shall have passed the following tests:

a. Examination of Product (see 4.5.1.1)

b. Operational Test (see 4.5.1.2)

c. Manufacturing Run-in Test (see 4.5.1.3)

4.5.1.1 Examination of product. Each control group shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each control group shall be operated long enough to permit the control group temperature to stabilize and to check characteristics and record data to assure that control group operation meets the requirements of this specification. This test shall be conducted at +25°C, -54°C and +71°C.

4.5.1.3 Manufacturing run-in test. Each control group shall be operated under the conditions specified below for a period of 10 hours without failure. A failure shall be specified as anything which causes malfunctioning of the control group. Only those adjustments shall be permitted which can be made by using controls and adjustments that are accessible to the operator during the normal use of the control group. This test shall be deleted if the reliability tests includes a test on each control group which consumes at least 10 hours of operation.

a. Temperature	Room ambient
b. Humidity	Room ambient
c. Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3g's

The control group shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10-hour period of operation. Where feasible, the control group shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship.

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Operation within the specified Limits of performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to that plane for five minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the 10-hour period of operation following the 10-minute vibration period, the control group shall be mechanically cycled periodically through its various phases of operation Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future control group. A record shall be kept of all failures (see 6.2.2). The 10-hour period specified above may be composed of several shorter periods to conform with standard working hours. This test is not required if a reliability assurance test is selected which includes a test on each control grOUp Which consumes at least 10 hours of operation.

4.5.2 Sampling tests. Control groups selected for sampling tests shall first have passed the individual tests. Control groups will be selected for sampling tests by the government inspector in accordance with table VIII.

TABLE VIII. Sampling tests sample selection

Quantity of Control Groups Offered for Acceptance	Quantity to be Selected for Sampling Test 1/
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

1/ Sampling Tests are not required when reliability assurance tests are conducted.

4.5.2.1 Scope of sampling tests. As a minimum, each control group selected for sampling tests shall be subjected to the following tests:

a. Complete operational tests at ambient room conditions, making all necessary measurements to assure that all requirements have been met.

b. Operational tests at certain environmental conditions. The conditions may vary for each control group tested and shall be based on results of the first article, individual, and special tests.

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c. Manufacturing run-in tests specified in 4.5.1.3 except that the test duration shall be 120 hours with no restrictions on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future control groups.

4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (sampling) phase shall be conducted.

4.5.3.1 Qualification phase. Prior to the acceptance of control groups under the contract or purchase order, a minimum of three control groups shall be tested in accordance with MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of control groups to be used shall be those listed in table V of MIL-STD-78l. For the qualification phase, Test Level E shall be used. The accept-reject criteria for Test Plan IV shall be used.

4.5.3.2 Reliability sampling phase tests. Samples of the control group shall be tested In accordance with MIL-S781 under the section entitled "Sampling Phase of Production Reliability Tests". For the sampling phase, Test Level E shall be used, except that the control group shall operate continuously. The accept-reject criteria for Test Plan V shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.5.3.2.1 Lot size for sampling phase. The control groups constructed during a two-month period shall be specified as one lot. The accept-reject criteria for Test Plan IV shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract of purchase order, no control groups shall be shipped until an accept decision is reached under each lot (see 6.2.1).

4.5.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be made available and accepted by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests (see 6.2.2).

4.5.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the control group and to assure quality control. The control group selected for special tests may be selected from control groups previously subjected to the sampling or reliability assurance tests.

4.5.4.1 Special tests schedule. Selection of control groups for special tests shall be made as follows:

a. On an early control group, after an engineering or material change.

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b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.5.4.2 Scope of special tests. Special tests shall consist of such tests as are acceptable to the procuring activity. Test procedures previously authorized for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare and make available a test procedure for procuring activity acceptance prior to conducting the tests (see 6.2.2).

4.5.5 Control group failure. Should a failure occur during either the samplings or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine If the failure is an isolated case or design defect.

C. Make available to the procuring activity, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests (see 6.2.2).

d. Where practical, include a test in the individual test to check all control groups for this requirement until assurance is obtained that the defect has been corrected.

4.5.6 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on control groups that have passed the individual tests. The life test shall be performed under the conditions specified 4.5.6.1. The life test sample will be selected by the government inspector in accordance with the following. (Control groups which have successfully passed the Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of control groups equal to, or more than, that listed in table IX receive at least 300 hours each of test time.

TABLE IX. Life test sample selection

Quantity of Control Groups Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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4.5.6.1 Test conditions. The life test shall be conducted under the following simulated service conditions:

a.	Temperature	Normal room
be	Altitude	Normal ground
c.	Humidity	Room ambient
d.	DC Voltage	5 ±0.1 volts dc, 28 ±0.1 volts dc, +4 ±0.01 volts dc, -4 ±0.01 volts dc

4.5.6.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the control group temperature to stabilize. Periodically, the control group shall be turned on and off several times and put through its various phases of operation.

4.5.6.3 Performance check. At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed in this specification shall be subject to acceptance by the procuring activity (see 6.2.2).

4.5.6.4 Test data. The contractor shall keep a daily record of the performance of the control group, making particular note of any deficiencies or failures (see 6.2.2). In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following

a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure

4.5.7.1 Failure report. In the event of a failure, the government inspector shall be notified immediately. A report shall be prepared and made available to the procuring activity upon completion of the test (see 6.2.2). In this report, the contractor shall propose suitable and design or materizd corrections for all failures which occurred (see 6.2.2). The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Electromagnetic interference (EMI) tests. EMI tests shall be performed in accordance with MIL-STD-462 to determine that the control group meets the requirements of MIL-STD-461 for Class A1 equipment (see 3.3.8).

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4.7 Environmental tests. The control group shall be subjected to the environmental tests specified in 4.7.1 through 4.7.1.6.

4.7.1 Vibration.

4.7.1.1 Random vibration test. The control group shall be subjected to the vibration spectrum shown in figure 5 for one hour in each of three mutually perpendicular axes. The control group shall be energized and shall meet all specification operational requirements during the random vibration tests.

4.7.1.2 Gunfire vibration test. The control group shall be subjected to the gunfire vibration requirements of figure 6 with the test duration consisting of eight six-minute intervals, each separated by 6-minutes with vibration removed (48 minutes total vibration exposure per axis). The control group need not be operating during the gunfire vibration tests, but shall operate as required after the completion of the gunfire vibration tests.

4.7.1.3 Random workmanship vibration test. The control group shall be subjected to the vibration spectrum shown in figure 5 for 30-minute test with the joystick oriented in the vertical axis and the converter oriented in the lateral axis as shown in figure 7. The control group need not be operating during the random workmanship vibration test, but shall operate as required after the completion of the random workmanship vibration tests.

4.7.2 Temperature-altitude. The control group shall be subjected to the temperature-altitude combination tests of MIL-E-5400 for Class 1 equipment.

4.7.3 Shock.

4.7.3.1 Operational. The control group shall be subjected to shock tests in accordance with MIL-E-5400 except the amplitude for the joystick shall be 20g in longitudinal axis, 22g in the lateral axis, and 35g in the vertical axis. The electronics package shall be subjected to amplitudes of 20g in longitudinal axis, 30g in the lateral axis, and 46g in the vertical axis. Figure 7 specifies the appropriate axes.

4.7.3.2 Crash safety. The control group shall be subjected to the crash safety shock tests specified in MIL-E-54U0 except that the amplitudes for the joystick shall be 40g in longitudinal axis, 40g in the lateral axis, and 47g in the vertical axis. The amplitudes for the electronic package shall be 40g in longitudinal axis, 40g in the lateral axis, and 55g in vertical axis

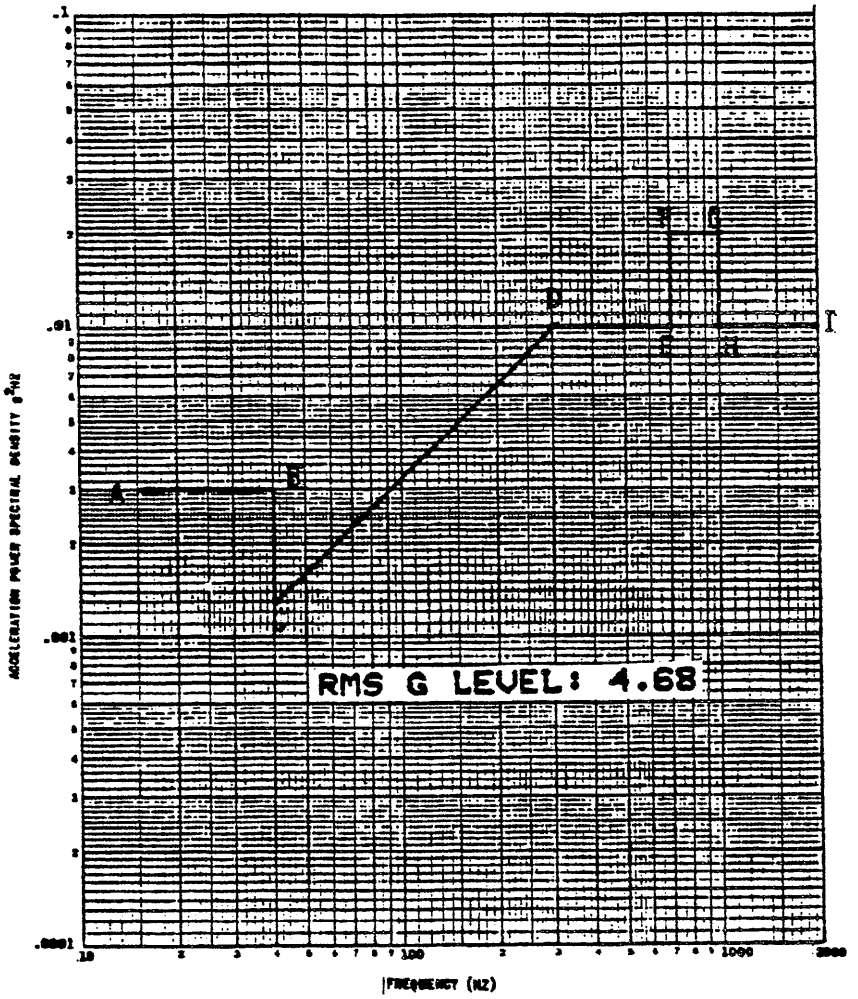
4.7.4 Humidity. The control group shall be subjected to the humidity tests in accordance with MIL-T-5422.

4.7.5 Salt spray. The control group shall be subjected to salt spray tests in accordance with MIL-T-5422.

4.7.6 Explosion The control group shall be subjected to explosion tests in accordance with MIL-T-5422.

4.8 Warm-up time. After stabilization at -54°C, the control group shall be energized for three minutes prior to recording operational test data.

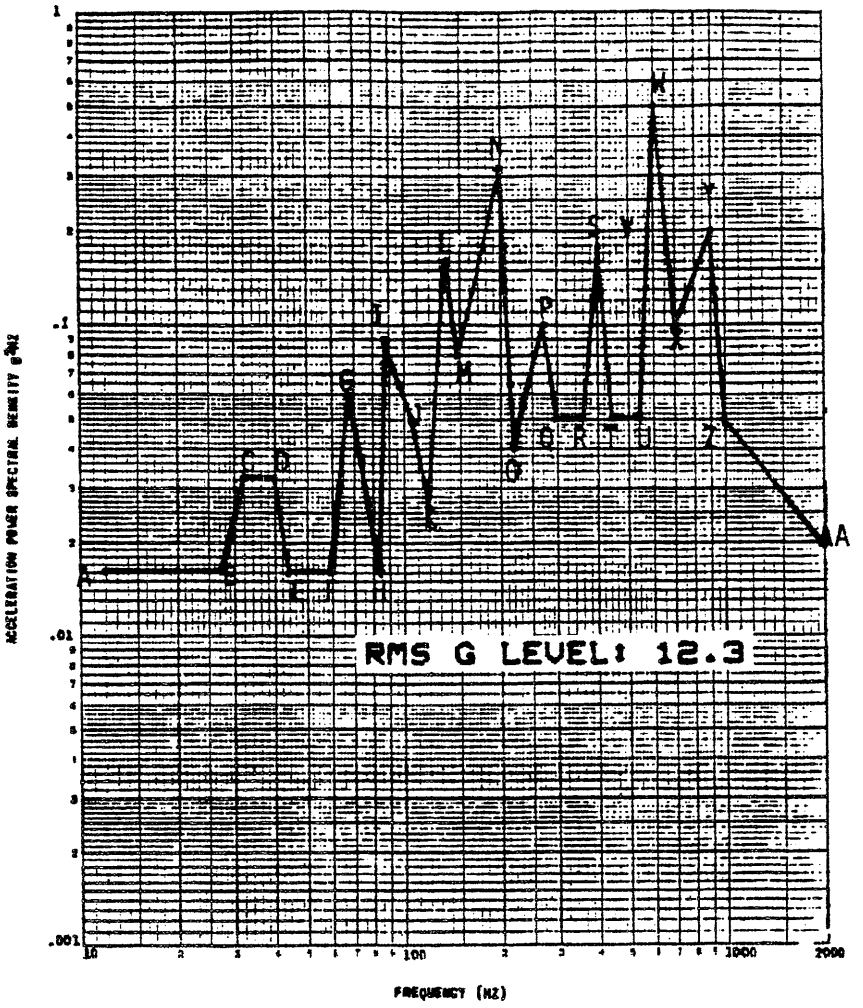
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Frequency (Hz)	g^2/Hz
16	.003
40	.003
44	.0013
300	.01
696	.01
700	.02
1000	.02
1004	.01
2000	.01

FIGURE 5. Operational (random) vibration spectrum

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Letter	Freq.(Hz)	ASD Value(g^2/Hz)	Letter	Freq.(Hz)	ASD Value(g^2/Hz)
A	12	.016	N	200	.32
B	28	.016	O	220	.04
C	32	.032	P	268	.1
D	40	.032	Q	300	.05
E	44	.016	R	360	.05
F	60	.016	S	400	.18
G	68	.06	T	440	.05
H	84	.016	U	540	.05
I	88	.09	V	560	.2
J	112	.05	W	600	.5
K	120	.026	X	700	.1
L	136	.16	Y	900	.2
M	148	.08	Z	1000	.05
			AA	2000	.02

FIGURE 6. Gunfire (random) vibration exposure

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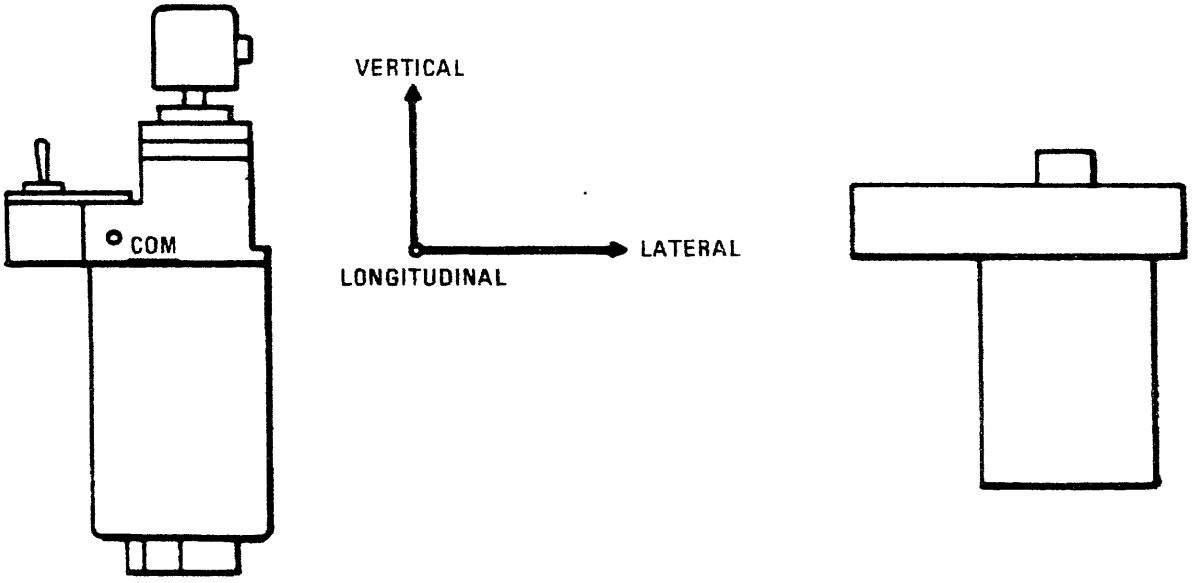


FIGURE 7. Guided Weapon Control Group OK-293A/AWW depicting appropriate vertical and longitudinal axes for shock tests.

4.9 Test procedures. The procedures used for conducting first article tests and quality conformance tests shall be prepared by the contractor and made available to the procuring activity (see 6.2.2). The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures shall be provided and may be used when their use is agreeable to the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1).

4.10 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of the specification and the contract.

4.11 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the rquirements of Section 5.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or industrial as specified in the contract (see 6.2.1).

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5.1.1 Level A.

5.1.1.1 Cleaning. Control groups shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Control groups shall be dried in accordance with MIL-P-116.

5.1.1.3 Unit packs. Unless otherwise specified in the contract, (see 6.2.1), each control group shall be individually unit packed in accordance with the methods of MIL-P-116 designated herein ensuring compliance with the applicable requirements of that specification.

5.1.2 Industrial. The industrial preservation of control groups shall be In accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A or industrial, as specified in the contract see 6.2.1).

5.2.1 Level A. The packaged control group shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, spectal requirements. In lieu of the closure and waterproofing requirements in the Appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, two tnches minimum width, conforming to PPP-T-60, Class 1 or PPP-T-76. Banding reinforcement requirements) shall be applied in accordance with the Appendix to PPP-B-636 using nonmetallic or tape banding only. When the gross weight exceeds 200 pounds or the container length and width is 48 inches x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

5.2.2 Industrial. The preserved control groups shall be packed in accordance with MIL-STD-1188.

5.2.3 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or purchase order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.3.1 Level A. Control groups, packed as specified in 5.2.1, shall be unitized on pallets accordance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.3 Marking.

5.3.1 Level A. In addition to any special or other identification marking specified in the contract (see 6.2.1), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Industrial. Industrial marking shall be in accordance with MIL-STD-1188.

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5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1 and 5.2.2) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.11.

6. NOTES

6.1 Intended use. The control group is airborne equipment and is intended to interface with the AN/AWW-7B and AN/AWW-9 Data Pods to provide cockpit control of the WALLEYE Data Link Guided Weapons when it becomes necessary to retarget or steer the weapon.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. If first article inspection is required (see 3.1, 6.3).

c. Additional individual tests, if required (see 4.5.1 and table VII).

d. Lot acceptance for each lot, if other than specified in 4.5.3.2.1.

e. Test procedure modifications or additional tests, if required (see 4.9).

f. Selection of applicable levels of packaging and packing (see 5.1).

g. Preservation and packing, Level A or industrial (see 5.1 and 5.3),

h. Unit packs, if other than specified in 5.1.1.3.

i. If special or other identification marking is required (see 5.3.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (N) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in

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accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4, 4.9	Procedure, First Article Inspection	DI-T-4901	
4.4.1, 4.4.3	Report, First Article Inspection	DI-T-4902	
4.5, 4.5.4.2, 4.5.5c, 4.9	Procedures, Production/Acceptance Inspection	DI-T-4903	
4.5, 4.5.6.3	Reports, Production Inspection	DI-T-49O4	
4.5.3.3	Procedures, Reliability Tests	DI-R-7035	
4.5.1.3, 4.5.6.4 4.5.7.1	Report, Failure	UDI-R-21141A	
4.5.7.1	Plan, Failure Data Collection, Analysis and Correction	UDI-T-237l9	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. H, AMSL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval publications and Forms Center or as directed by the contracting office.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one control group. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the documents' first article.

6.4 Associated equipment. The control group should be designed to operate with the associated equipment:

a. Guided Weapon Control - Monitor Sets AN/AWW-7B and AN/AWW-9

b. Computer Set, Tactical AN/ASN-91.

Preparing Activity
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(Project No. 1430-N110)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-85231 (AS)	Guided Weapon Control Group OK-293A/AWW

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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MILITARY SPECIFICATION

GUIDED WEAPON CONTROL
GROUP OK-293A/AWW

This amendment forms a part of Military Specification MIL-G-85231(AS) dated 30 October 1984, and is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department Of Defense.

Page 7

Delete “3.4.a” in its entirety and substitute;

a.. Under standard conditions (see 3.3.10), the control group shall provide the regulated outputs as specified in Table II and the Appendix.

Page 30

Add "Appendix":

APPENDIX
DATA WORD REQUIREMENTS FOR
GUIDED WEAPON CONTROL GROUP 0K-0293A/AWW

10. SCOPE.

10.1 Scope. This appendix details the data word requirments of the Guided Weapon Control Group OK-293A/AWW. This appendix is a mandatory part of this specification. The information contained herein is intended for conpliance.

20. APPLICABLE DOCUMENTS

20.1 This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 16-BIT digitally encoded word. The control group shall use a 16-BIT digitally encoded word tormat as shown below:

SYNC	A N T	S E L	LSB	2ND	MSB	8 SPARES	LAST TWO
1	2	3	4	5	6	7-14	15-16

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Where LSB is “Least Significant Bit” and MSB is “Most Significant Bit, ” the aft antenna shall be selected if the third bit is a “one”; and the forward antenna shall be selected if the third bit is a “zero. Eight spares shall be provided for future growth. The last two bits shall be hardwired to a “zero.”

30.2 Data. The data shall consist of a 16-bit parallel word with two bits “Sync” and 12 bits of “Data” with the last two bits tied to “zero.” The first half of each bit shall be “Sync” with the second half of each blt data being a “one” or “zero”. For “Sync,” the second half of each bit is “Sync.” The clock rate shall be 50 +4.4, -4.0 Hz.

30.3 Clock rate. The clock rate shall be determined by the 1300 Hz voltage controlled oscillator (VCO). The VCO sine wave output shall be converted to a square wave and then divided by 26 to obtain a 50 percent duty cycle. During standard conditions the clock rate is as follows:

a. $\frac{1300 \pm 5\text{Hz}}{26}$ or 50 ± 0.2 Hz for a “Sync.”

b. $\frac{1202 \pm 5\text{Hz}}{26}$ or 46.23 ± 0.2 Hz for a “Zero.”

c. $\frac{1398 \pm 5\text{Hz}}{26}$ or 53.77 ± 0.2 Hz for a “One.”

During service conditions the clock rate is as follows:

a. $\frac{1300 \pm 16\text{Hz}}{26}$ or 50M.62 Hz for a “Sync.”

b. $\frac{1202 \pm 16\text{Hz}}{26}$ or 46.23 ± 0.62 Hz for a “Zero.”

c. $\frac{1398 \pm 16\text{Hz}}{26}$ or 53.77 ± 0.62 Hz for a “One.”

30.4 Data definition (standard and service conditions). The data shall be defined for standard and service conditions as follows:

a. Standard conditions

(1) A sync bit shall be defined as 1269 to 1331 Hz.
(2) A data zero shall be defined as 1197 to 1250 Hz.
(3) A data one shall be defined as 1351 to 1403 Hz.

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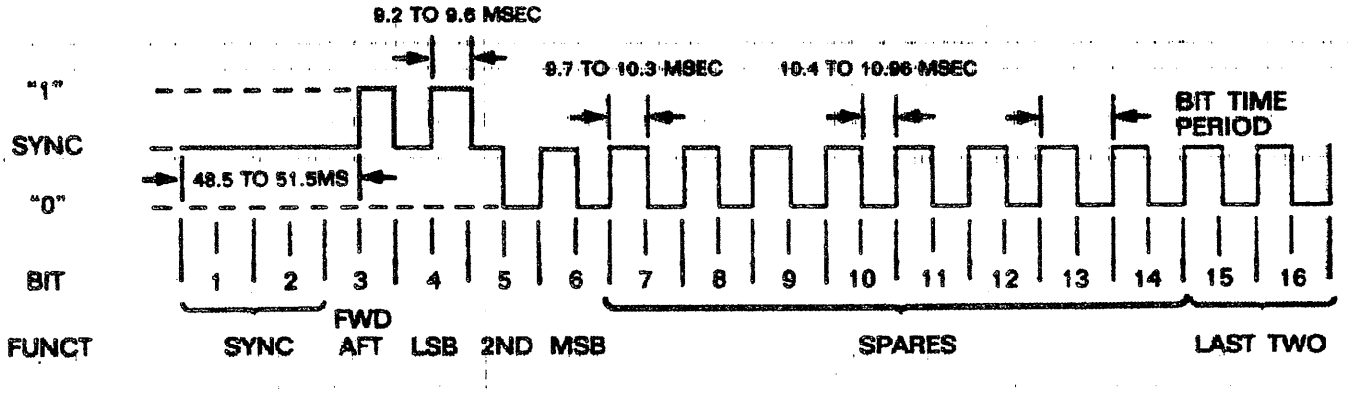
b. Service conditions

(1) A Sync bit shall be defined as 1258 to 1342 Hz
(2) A data zero shall be defined as 1186 to 1250 Hz.
(3) A data one shall be defined as 1351 to 1414 Hz.

30.5 Word format. An example of the word format is specified in figure 7 and as follows:

a. Aft. Ant. shall be a one, the next data bit shall be a one, and all other data bits shall be a zero. The last two data bits, 15 and 16, shall be tied to zero permanently.

30.6 Format logic. Format logic, shall be as specified in Table X for presently assigned weapons.



NOTES:

1. The 1 level shall be +4.7 to +5.3 volts dc.
2. The SYNC level shall be +2.35 to +2.65 volts dc.
3. The 0 level shall be 0.0 to +0.30 volts dc.

FIGURE 7. Modulation waveform

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TABLE X. Format logic

Channel Select Switch Letter	FWD AFT	LSB	2ND	MSB	Spares 7-14	Last Two	Channel	Format
A	0 or 1	1	0	0	0	0	A	Phase I
C	0 or 1	0	1	0	0	0	C	Phase I
E	0 or 1	1	1	0	0	0	E	Phase I
F	0 or 1	0	0	1	0	0	A	Phase II
H	0 or 1	1	0	1	0	0	C	Phase II
J	0 or 1	0	1	1	0	0	E	Phase II
K	0 or 1	1	1	1	0	0	None	Future
M	0 or 1	0	0	0	0	0	Manual	For AWW-7 Pod

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